

Overview

This project explores transcriptomic heterogeneity in pancreatic ductal adenocarcinoma (PDAC) using RNA-seq data from tumor center (TC) and tumor periphery (TP) regions. The analyses focused on identifying molecular subtypes, investigating the impact of neoadjuvant chemotherapy, and characterizing the expression of the candidate marker **CYP3A5**.

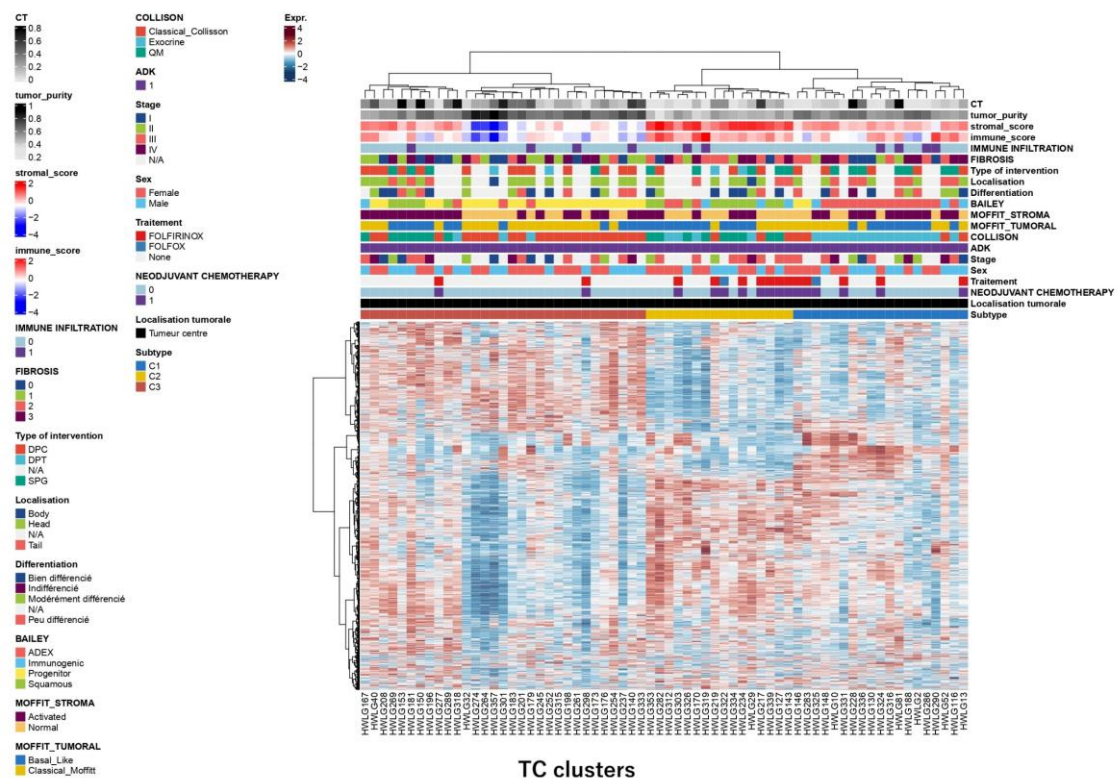
The work was conducted as part of a research internship in translational cancer bioinformatics.

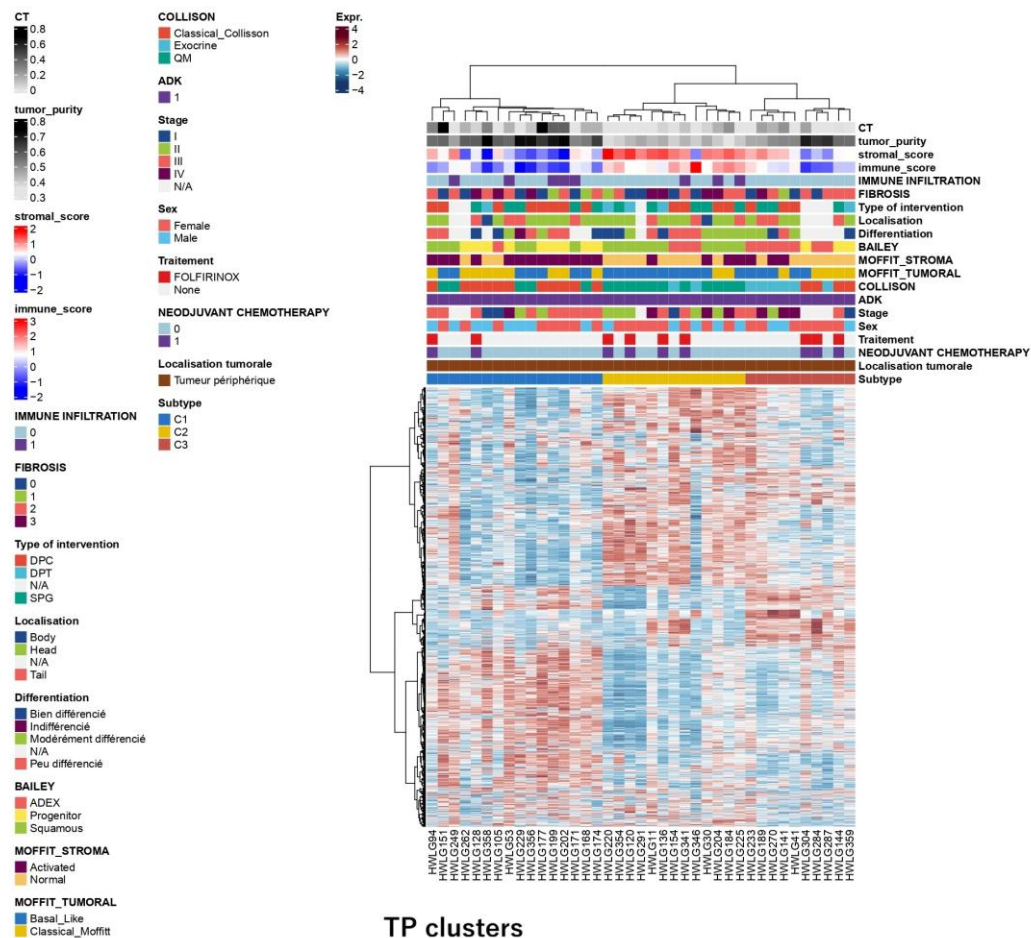
Cohort Description

The cohort included:

- **66 Tumor Center (TC) samples**
- **39 Tumor Periphery (TP) samples**

After quality control and removal of duplicated samples, transcriptomic analyses were performed independently in TC and TP regions.





Methods

Transcriptomic Clustering

- Selection of the **top 10% most variable genes**
- Consensus clustering performed independently in TC and TP samples
- Identification of **3 transcriptomic clusters** per region

Functional Characterization

- Gene Set Enrichment Analysis (GSEA)
- Hallmark pathway enrichment
- Biological interpretation of transcriptomic subtypes

Statistical Analyses

- Association between clusters and chemotherapy exposure using Fisher's exact test
- Survival analyses using Kaplan–Meier curves and log-rank tests
- CYP3A5 expression stratification using upper quartile threshold
- Exploratory cutoff optimization using maximally selected rank statistics

Results

Tumor Center (TC) Clustering

Three transcriptomic subtypes were identified:

Cluster C1 (n = 19)

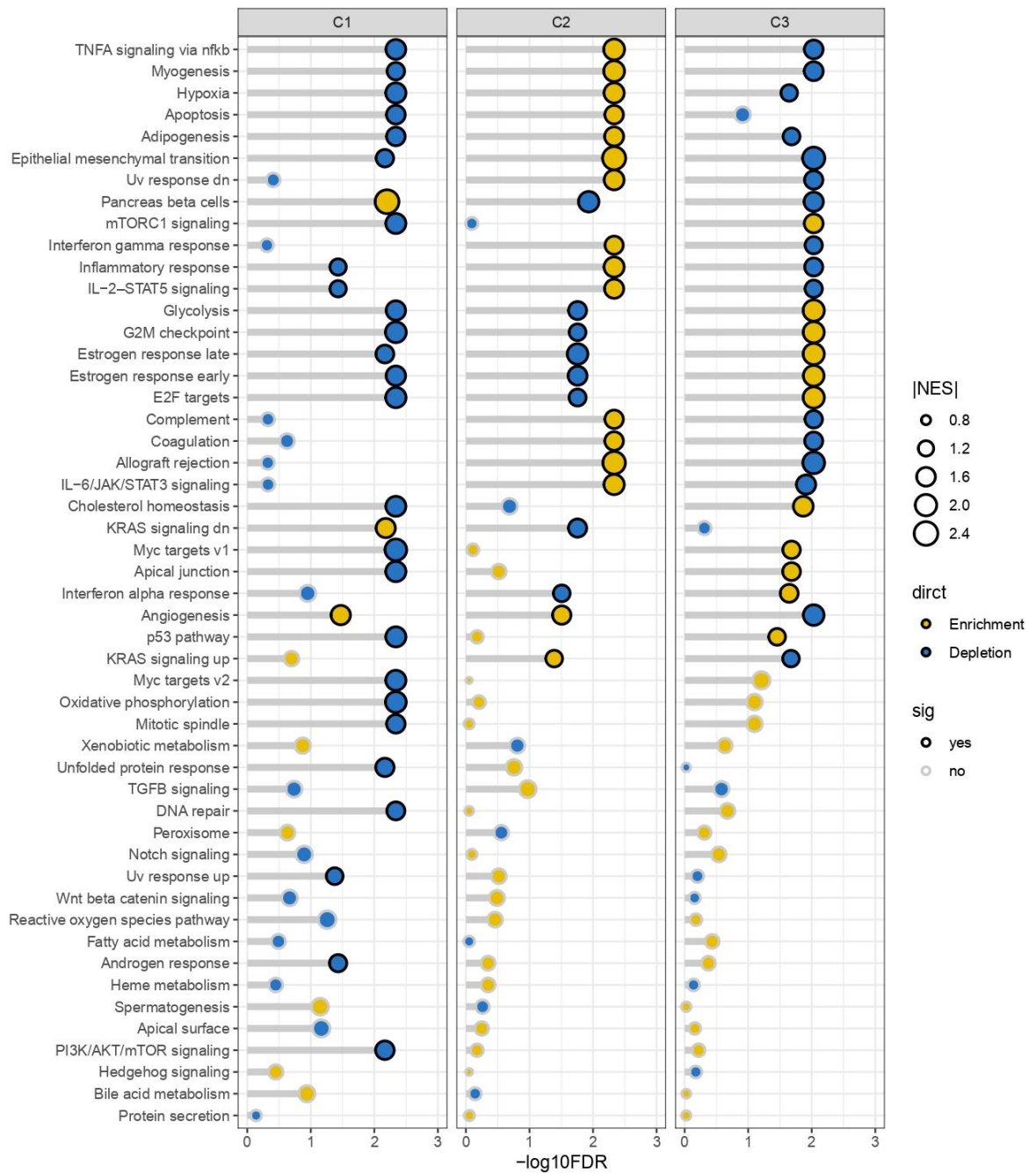
- Immune and stromal-associated phenotype
- Enrichment in inflammatory pathways
- Reduced proliferation signatures

Cluster C2 (n = 16)

- EMT and stromal activation phenotype
- Enrichment in:
 - TNF α /NF κ B signaling
 - IL6/JAK/STAT3 pathways
 - inflammatory response

Cluster C3 (n = 31)

- Highly proliferative tumor phenotype
- Enrichment in:
 - glycolysis
 - MYC targets
 - E2F signaling
 - G2M checkpoint
 - cell cycle pathways



Tumor Periphery (TP) Clustering

Three distinct TP subtypes were also identified, highlighting spatial transcriptomic heterogeneity between tumor regions.

- One subtype showed a more differentiated phenotype
- One subtype was associated with aggressive molecular features
- One subtype displayed activated stromal signatures

Association with Neoadjuvant Chemotherapy

Tumor Center (TC)

A significant association was observed between transcriptomic clusters and chemotherapy exposure:

- $p = 0.0024$

Distribution

- C1: 6 treated / 13 untreated
- C2: 8 treated / 8 untreated
- C3: 2 treated / 29 untreated

Interpretation

The proliferative cluster (C3) was predominantly composed of untreated tumors, whereas treated tumors were more frequently distributed across C1 and C2.

These findings suggest that chemotherapy may reduce highly proliferative tumor states and promote stromal or inflammatory remodeling of the tumor microenvironment.

Tumor Periphery (TP)

No significant association was observed between TP clusters and chemotherapy exposure:

- $p = 0.44$

This suggests that chemotherapy-associated transcriptomic changes are more detectable in the tumor center than in peripheral tumor regions.

Survival Analysis

Survival analyses were performed in both TC and TP transcriptomic subtypes.

Tumor Center (TC)

No statistically significant differences in overall survival were observed across TC transcriptomic subtypes:

- **log-rank p = 0.12**

Tumor Periphery (TP)

Similarly, no significant association between TP transcriptomic subtypes and survival was observed.

The absence of statistically significant differences may be related to the limited sample size of the cohort.

CYP3A5 Expression Analysis

CYP3A5 expression was stratified into **High** and **Low** groups using the upper quartile threshold (top 25%), consistent with previous Pancréas-CGE analyses.

Association with Survival

Disease-specific survival could not be evaluated because the available clinical annotations did not contain cause-of-death information. Therefore, the association between CYP3A5 expression and progression risk was assessed using progression-free survival (PFS).

Predefined Cutoff Analysis

Using the predefined upper-quartile cutoff, no statistically significant association between CYP3A5 expression and progression-free survival was observed in either TC or TP samples.

Exploratory Cutoff Optimization

An exploratory outcome-driven cutoff optimization was then performed using maximally selected rank statistics.

Although the optimized cutoffs slightly improved the separation of survival curves, the results remained non-significant in both tumor regions.

Overall, both approaches led to the same conclusion:

No significant association between CYP3A5 expression and progression risk was observed in this cohort.

Biological Associations of CYP3A5

Despite the absence of prognostic significance, CYP3A5 expression showed strong and consistent associations with transcriptomic programs in both tumor center and tumor periphery.

Global correlation and enrichment analyses demonstrated that:

- High CYP3A5 expression was associated with:
 - epithelial differentiation programs
 - metabolic pathways
- Low CYP3A5 expression was associated with:
 - stromal signatures
 - inflammatory pathways
 - hypoxia-related programs
 - mesenchymal features

These findings suggest that CYP3A5 is more strongly associated with intrinsic tumor biology and differentiation states than with chemotherapy exposure or progression risk.

The results were summarized using mirror dotplots for both TC and TP regions.